

values; ii) in the strongly underdamped case, the second-order phase transition turns into a first-order phase transition (note the jump in the curve for $m = 10$), which is in line with what has been reported before [26, 27].

In any case, for all inspected values of the mass, the asynchronous state ($r \ll 1$) always exists over a finite range of mean coupling strengths. In this parameter region the temporal statistics of the dynamics is independent of K . In the remainder of the study, we will set $K = 0$.

B. Power Spectrum Calculation

To further analyze the fluctuations in the incoherent regime, we compute the power spectrum of individual oscillators, denoted as $S_{x_\ell}(\omega)$, where $x_\ell = e^{i\theta_\ell}$. The power spectrum is defined as:

$$S_{x_\ell}(\omega) = \lim_{T \rightarrow \infty} \frac{1}{T} \langle |\tilde{x}_\ell(\omega)|^2 \rangle, \quad \tilde{x}_\ell(\omega) = \int_0^T e^{i\omega t} x_\ell(t) dt, \quad (5)$$

where $\tilde{x}_\ell(\omega)$ is the Fourier transform of $x_\ell(t)$ computed over a finite time interval T . Common temporal statistics are characterized by the averaged power spectrum,

$$S_z(\omega) = \frac{1}{N} \sum_{\ell=1}^N S_{x_\ell}(\omega). \quad (6)$$

We obtain these spectra by integrating the full set of second-order differential equations (Eq. (1)) using a fourth-order Runge-Kutta method with a relative tolerance of 10^{-8} . We refer to this direct numerical integration as the network dynamics (ND) method and use it as a benchmark against the iterative mean-field (IMF) method introduced in the following section. In Fig. 2 and Fig. 3, the ND results are shown as black dashed lines for comparison with the IMF method.

The simulation is run for a total time period T (with $T = 10^5$ for Fig. 2 and $T = 10^4$ otherwise), discarding the initial time $t_d = 1000$ to eliminate transients. To allow oscillator-resolved analysis, the natural frequencies ω_ℓ are drawn from the prescribed distribution and held fixed, as is the Gaussian-distributed connectivity matrix $K_{\ell m}$. The system size is chosen as $N = 10^4$ to minimize dependence on the specific realization of the connectivity matrix $K_{\ell m}$, which is also drawn from a Gaussian distribution.

The power spectrum is computed using the fast Fourier transform (FFT). The single-oscillator power spectrum $S_{x_\ell}(\omega)$ is evaluated for a selected oscillator ($\ell = 3$ in our case), while the mean power spectrum $S_z(\omega)$ is obtained by averaging over all oscillators:

$$S_z(\omega) = \frac{1}{N} \sum_{\ell=1}^N |\tilde{x}_\ell(\omega)|^2. \quad (7)$$

The results are smoothed by binning neighboring frequency points to improve spectral accuracy. We divided

the frequency axis into $M = 2000$ uniform bins, with $T/(Mdt)$ frequency points averaged per bin using an accumulation method to reduce statistical noise.

III. ITERATIVE MEAN FIELD METHOD

In this study, we employ the iterative mean field (IMF) method, often also referred to as dynamic mean field method [28], to analyze the asynchronous states of the Kuramoto model with inertia and disorder in coupling and natural frequencies. This method provides a self-consistent approach to compute the power spectra of single oscillators, approximating the network noise experienced by each oscillator as a Gaussian process, iteratively refined until convergence. The method has been used for spin systems in the early 1990's [29], for networks of spiking neurons [19–21], and specifically for the Kuramoto model with disorder in the coupling coefficients (but without inertia) in [4–6, 30].

For our system with inertia, we start from the equations of motion for the oscillators Eq. (1) and express the coupling term in the dynamics as a multiplicative network noise:

$$m\ddot{\theta}_\ell(t) + \dot{\theta}_\ell(t) = \omega_\ell + \text{Im} \left(e^{-i\theta_\ell(t)} \zeta_\ell(t) \right), \quad (8)$$

where the network noise $\zeta_\ell(t)$ is defined as:

$$\zeta_\ell(t) = \sum_{m=1}^N K_{\ell m} e^{i\theta_m(t)}. \quad (9)$$

For large networks in the asynchronous regime (excluding the Volcano transition case observed with symmetric coupling coefficients, see [30, 31]), this noise behaves as a superposition of independent Gaussian processes due to the central limit theorem. The autocorrelation of the network noise can be written as:

$$\begin{aligned} \langle \zeta_\ell(t) \zeta_\ell(t') \rangle &= \left\langle K_{\ell m} K_{\ell n} e^{i(\theta_m(t') - \theta_n(t))} \right\rangle \\ &= \sum_{m,n} \left(\frac{k^2}{N} \delta_{mn} + \frac{K^2}{N^2} \right) \left\langle e^{i(\theta_m(t') - \theta_n(t))} \right\rangle. \end{aligned} \quad (10)$$

Assuming uncorrelated oscillators yields

$$\left\langle e^{i(\theta_m(t') - \theta_n(t))} \right\rangle = \delta_{mn} \left\langle e^{i(\theta_m(t') - \theta_m(t))} \right\rangle, \quad (11)$$

leading to the simplified form

$$\langle \zeta(t) \zeta(t') \rangle = (k^2 + K^2/N) \left\langle e^{i(\theta(t') - \theta(t))} \right\rangle. \quad (12)$$

The right-hand side thus represents the autocorrelation of a single oscillator's phase pointer. In the Fourier domain, the network noise spectrum $S_\zeta(\omega)$ is proportional to the power spectrum of a single oscillator averaged over the network:

$$S_\zeta(\omega) = (k^2 + K^2/N) S_z(\omega), \quad (13)$$